Assignment 1: Reinforcement Learning

The goal of this assignment is to get you familiar with writing python code to describe and solve RL problems pertain to robot simulation. The hand-in for each group could be a PDF file or jupyter notebook containing texts, figures, code and results.

1 Warm-up exercise

Install following python libraries. Remember to create virtual/conda/mamba environment if you don't want to overwrite the existing dependencies:

- Gymnasium; note do not install openai-gym which is a deprecated library for the same purpose
- Mujoco
- Reinforcement learning libraries such as stable baselines 3

Check and run the minimal 'hello world' examples of these libraries according to their documentation. Read the gymnasium code defining the "HalfCheetah" environment (e.g. gymnasium/envs/mujoco/half_cheetah_v4.py) and answer following questions:

- What are the state and action spaces for this RL environment?
- Is the initial state of the environment randomized? If yes, what probabilistic distribution is used?
- How is the reward function defined to incentivize a moving forward behaviour?

2 Train the first RL agent

Perform following tasks:

- Write a small code snippet to call an RL algorithm (e.g. PPO) to train the HalfCheetah environment. **Hint:** the agent is likely to start learning with at least 10^6 and it will be 1-3 minutes or even less on modern computers.
- Plot how the episodic return improves as the training proceeds. **Hint:** for stable baselines 3, check environment wrapper *Monitor*.

- Check how the HalfCheetah runs with the trained policy and post a few images illustrating the learned behaviour. **Hint:** you might check env.render() function for rendering the simulation environment.
- Is the trained behaviour as expected? Explain why such a behaviour emerges given the reward design.

3 Change the behaviour

Now try to change the reward design to learn another behaviour, e.g. moving backwardly. How would you do that? **Hint:** you can define a new environment deriving the HalfCheetahEnv class and override the function about reward calculation. Train the new environment with RL again and create images showing the new target behaviour.

4 Change the simulation model

Give a read for the basics about defining a simulation model with the mujoco xml file, especially on the definition of body and geom attributes:

`https://mujoco.readthedocs.io/en/stable/XMLreference.html`

and the model definition for the HalfCheetah robot:

`https://github.com/Farama-Foundation/Gymnasium/blob/main/gymnasium/envs/mujoco/assets/half_cheetah.xml`

Make a change to the physical parameters of the HalfCheetah, e.g. elongating the torso or a thigh. Visualize the changed model with images and report if the trained policies above still work on the modified HalfCheetah.